Chiral quantum walks

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0) CLASSIFICATION: network geometry influences TRS, which is interesting for inducing directionality

1) network-based view of TR

2) controlling TA

3) state transport

Given its importance to many other areas of physics, from condensed-matter physics to thermodynamics, time-reversal symmetry has had relatively little influence on quantum information science. Here we develop a network-based picture of time-reversal theory, classifying Hamiltonians and quantum circuits as time symmetric or not in terms of the elements and geometries of their underlying networks. Many of the typical circuits of quantum information science are found to exhibit time asymmetry. Moreover, we show that time asymmetry in circuits can be controlled using local gates only and can simulate time asymmetry in Hamiltonian evolution. We experimentally implement a fundamental example in which controlled time-reversal asymmetry in a palindromic quantum circuit leads to near-perfect transport. Our results pave the way for using time-symmetry breaking to control coherent transport and imply that time asymmetry represents an omnipresent yet poorly understood effect in quantum information science.

"pave the way"

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I. INTRODUCTION

1) control over transfer

Controlling probability transfer in quantum systems is a central challenge faced in several emerging quantum technologies [1–5]. Here we develop an approach based on a complex network theory viewpoint of quantum systems [6–9], in which probability transfer is directed by the controlled breaking of time-reversal symmetry, creating a so-called *chiral quantum walk* [10–15].

2) breaking symmetry on purpose

The practical importance of time-reversal symmetry breaking stems from the fact that it is equivalent to introducing biased probability flow into a quantum system. It thus enables directed state transfer without requiring a biased (or nonlocal) distribution in the initial states or coupling to an environment [10,16–18].

3) introduce a bias = directionality

4) asymm. cond.

This work establishes, in terms of the geometry and edge weights (or gates) of the underlying graph, conditions on what makes a Hamiltonian (or circuit) time asymmetric. As well as allowing us to classify the time symmetry of well-known Hamiltonians and circuits, this knowledge allows us to develop active methods to break time symmetry and bias probability transfer, including using circuits to simulate time-symmetry-breaking Hamiltonians.

5) actively breaking symm.

As a demonstration, the most basic time-asymmetric process is identified and realized experimentally using room-temperature liquid-state nuclear magnetic resonance (NMR) on a 3-qubit system. We show that time asymmetry and thus

experimental realization

limited actions required

biased probability transport can be controlled with limited access to the system, namely by using local z rotations paired with a naturally occurring (or in our case, emulated) time-symmetric evolution. Through symmetry breaking we achieve state transfer probabilities approaching unity.

Since their recent introduction [10], continuous time chiral quantum walks have been studied in the context of energy transport in ultracold atoms and molecules [11], in nonequilibrium physics [12,13], as a tool for quantum search [14], and as a method to achieve near-perfect state transfer [10,15]. Our theory extends the contemporary analysis of time asymmetry such that it can now apply to gate sequences—of which the existing theory of time symmetry becomes a special case—and presents a network classification of the effect. Our experiments illustrate how active time-reversal symmetry breaking can be utilized with existing quantum technologies in the presence of limited control.

useful

this study

II. CHIRAL QUANTUM WALKS

Consider a unitary propagator U acting on a state space in which we have a preferred basis $\{|i\rangle\}$. We view U as performing a quantum walk over nodes labeled by i, and $U^\dagger$ as performing the time-reversed walk. There are two ways in which the walk U could be time symmetric. 2 symmetries possible

Amplitude time-reversal symmetry establishes a relationship $\langle i|U|j\rangle = U_{ij} = (U^\dagger)^*_{ij}$ between the internode transition amplitudes going forward and backward in time (where * denotes complex conjugation and † denotes Hermitian conjugation). Equivalently, it establishes a relationship $U_{ij} = U_{ji}$ U = U^T

ATS

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between opposing transition amplitudes in the same time direction.

PTS

It further implies what we call *probability time-reversal symmetry* (PTS): symmetry of the internode transition probabilities going forward and backward in time, $|U_{ij}|^2 = |(U^\dagger)_{ij}|^2$. This is equivalent to a lack of directional bias in the transport between any pair of nodes in the same time direction: $|U_{ij}|^2 = |U_{ji}|^2$. changing the square modulus

Breaking amplitude time-reversal symmetry is a necessary, but not sufficient, condition for breaking probability time-reversal symmetry. In studying internode transport, PTS is the relevant feature. It leads to a richer classification of processes and is the focus of this work. focus on PTS

III. HAMILTONIAN EVOLUTIONS

1) anything can be reduced to some graph structure

We first consider a unitary propagator $U = e^{-iHt}$ generated by some time-independent Hamiltonian H. In our network-based picture, with nodes labeled by i, we view the matrix elements $\langle i|H|j\rangle = H_{ij} = h_{ij}e^{i\alpha_{ij}}$ representing Hamiltonian H as forming a complex Hermitian adjacency matrix. Here h_{ij} and α_{ij} take only real values. For a nonzero value H_{ij}, two nodes i and j are said to be connected by an edge $e = (i,j)$, and the complex valued edge weight is given by the value of H_{ij}. A special case is an edge that connects i to itself, called a self-edge. Since it represents the expectation value of the energy in state $|i\rangle$, the weight H_{ii} of a self-edge must take a real value. Furthermore, self-edges are only needed if states $|i\rangle$ have different energies H_{ii}; otherwise they may be omitted without changing the underlying physics. Taken together, the nodes and edges define a support graph, corresponding to the Hamiltonian H, that we will use to classify the time-symmetry of U.

energy meaning of loops (regular graphs)

We begin by noting that if $\alpha_{ij} = 0$, i.e., all edge weights are real, then both amplitude and probability time-reversal symmetry hold. We will now search for other probability time-symmetric Hamiltonians.

A large class of such Hamiltonians are obtained by considering mapping $H \mapsto \Lambda^\dagger H \Lambda = H'$ (or equivalently $U \mapsto \Lambda^\dagger U \Lambda = U'$), where Λ is a diagonal unitary. Such a mapping will in general affect amplitude time symmetry. However, it cannot affect transition probabilities as $|U'_{ij}|^2 = |\langle i|e^{-i\Lambda^\dagger H \Lambda t}|j\rangle|^2 = |\langle i|\Lambda^\dagger e^{-iHt}\Lambda|j\rangle|^2 = |U_{ij}|^2$, and hence cannot affect probability time asymmetry. Thus, we will call these mappings *quasi-gauge-symmetry transformations*.

All Hamiltonians that are obtained from a Hamiltonian with real edge weights by such gauge transformations are thus probability time symmetric. To give an example, if the graph underlying H is a tree (of which a linear chain is a special case), there always exists [10] a Λ which removes all phases from the edge weights: $H_{ij} = h_{ij}e^{i\alpha_{ij}} \mapsto h_{ij}$. Such Hamiltonians hence never break probability time symmetry. This gives us our first class of time-symmetric evolutions: those generated by Hamiltonians whose internode couplings form a tree structure (this may include self-edges).

The class of probability time-symmetric networks is richer than those obtained by gauge transformations from real Hamiltonians. To find the other members of this class, we must consider the interference between walks along different paths. We start by breaking the evolution into commuting even and odd functions of H: $U = e^{-iHt} = \cosh(iHt) - \sinh(iHt)$. The probability time-symmetry condition $|U_{ij}|^2 = |U_{ji}|^2$ can now be expressed as

from |U_ij|=|U_ji|

$$\sinh(iHt)_{ji}\cosh(iHt)_{ij} = \sinh(iHt)_{ij}\cosh(iHt)_{ji}. \quad (1)$$

The physical interpretation of Eq. (1) is clear: $\sinh(iHt)_{ij}$ corresponds to transitions along paths of odd length, and $\cosh(iHt)_{ij}$ corresponds to transitions along paths of even length. Together these terms account for all possible paths between i and j [19]. A path from i to itself is called a cycle. parity meaning

5) bipartite graphs: one of the terms of (1) is always zero!

For graph geometries where between each pair of nodes (i,j) there are only exclusively even or exclusively odd paths (equivalent to there being no odd-length cycles in the graph), probability time symmetry must always hold as Eq. (1) is always satisfied. Such graphs are called bipartite, where the nodes can be partitioned to two disjoint sets and nonzero edge weights $H_{ij} \neq 0$ only connect nodes of different sets [20]. Note that this disallows self-edges, since such edges could be used to make cycles of arbitrary length. Bipartite graphs can also be understood in terms of gauge transformations, as their structure implies the existence of a gauge transformation $H \mapsto \Lambda^\dagger H \Lambda = -H$, which immediately implies the probability time-symmetry condition $|U_{ij}|^2 = |(U^\dagger)_{ij}|^2$. gauge transf. PTS

This leaves us with two overlapping classes of time-symmetric network geometries, trees with self-edges and bipartite graphs, where the overlap is the class of trees without self-edges or, equivalently, bipartite graphs without simple cycles. The remaining graph geometries are all potentially time asymmetric, with the degree of asymmetry determined by the values assigned to the edge weights. These findings are summarized with examples in Table I. SUMMARY

2) REAL values only

other classes

in ketbra notation, it just changes the phase...

PTS is safe

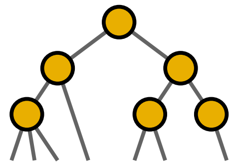
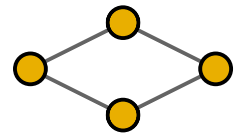
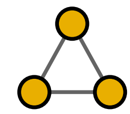
3) quasi-gauge

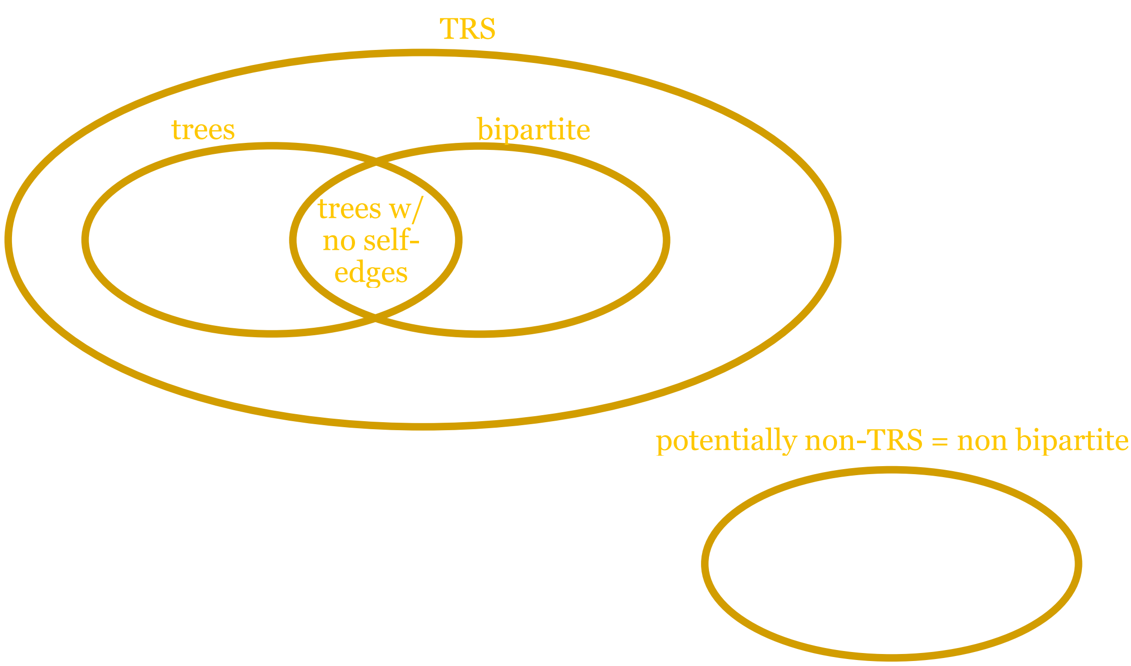
e.g. tree

it is possible to cancel out phases, so H becomes real, thus also symmetric

4) interference btw paths of diff. length: cases in which phases cannot be removed

TABLE I. In which network geometries do transition probabilities depend on the complex phases α_{ij} of the edges of the (effective Hamiltonian's) internode coupling graph? We are interested in how the transition probabilities in the site basis depend on α_{ij} and if certain values of the α_{ij} can break time-reversal symmetry.

	Linear chains	Trees (possibly with self-edges)	Bipartite graphs (only even cycles)	Non-bipartite graphs (some odd cycles)
	 particular tree...			
Probability time symmetric ($\forall\ \alpha_{ij}$)?	Yes	Yes	it changes just the sign Yes	No
Probability depends on α_{ij}?	No	No	Yes	Yes



Two important properties of the model now follow: (i) phases on tree graphs can be transformed out completely and (ii) the sum of phases along loops is invariant under gauge transformations.

The first property is illustrated in Figure 4. Let us take an arbitrary tree graph and pick a vertex m with only one neighbour. Redraw every other vertex on successive levels characterized by the distance d of the vertexes from the given vertex m. Note that the number of edges connecting two vertices, d, is by definition, unique in tree graphs. In such an arrangement only one edge emanates downwards from a given vertex on a line of $d > 0$ so Figure 4 represents the general neighbourhood of a vertex n having distance $d = 1$ from m. The indicated gauge transformation with $\alpha_n = -\theta$ removes the phase from the bottom edge. Then, one iterates the procedure for all vertices at level $d = 2$ and consecutively for all levels. In this way, all phases are removed. For the second property, pick an orientation on a loop of N vertexes and compute $\theta := \sum_{i=1}^{N} \phi_{i,i+1}$, considering $\phi_{N,N+1} \equiv \phi_{N,1}$. A gauge transformation $|n\rangle \mapsto e^{i\alpha_n}|n\rangle$, according to Eq. (9) leads to:

$$\phi_{n,n+1} \mapsto \phi_{n,n+1} + \alpha_n, \quad \phi_{n-1,n} \mapsto \phi_{n-1,n} - \alpha_n,$$

so the sum θ remains unaffected.

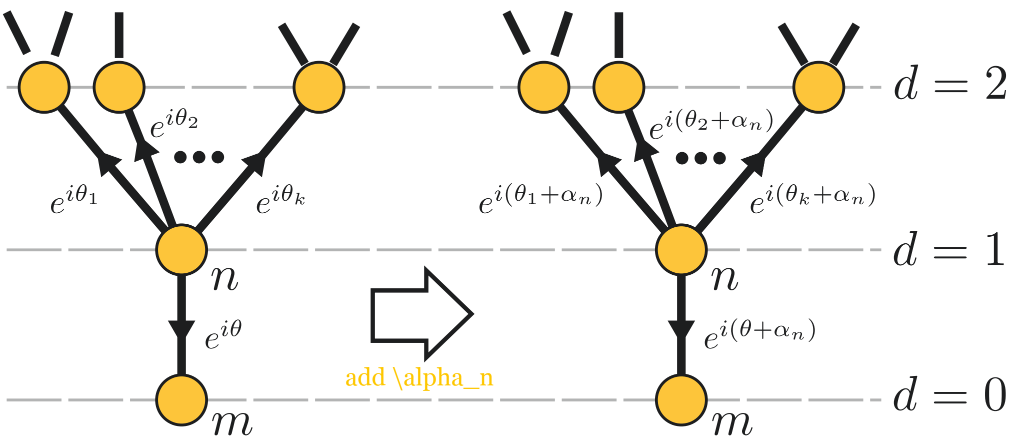


Figure 4 | Effect of the gauge transformation $|n\rangle \to e^{i\alpha_n}|n\rangle$ on vertex n. Phases on edges can be gauge-transformed without changing the transition amplitudes, as described in the text. Here we arrange the graph as a tree rooted at $d = 0$.

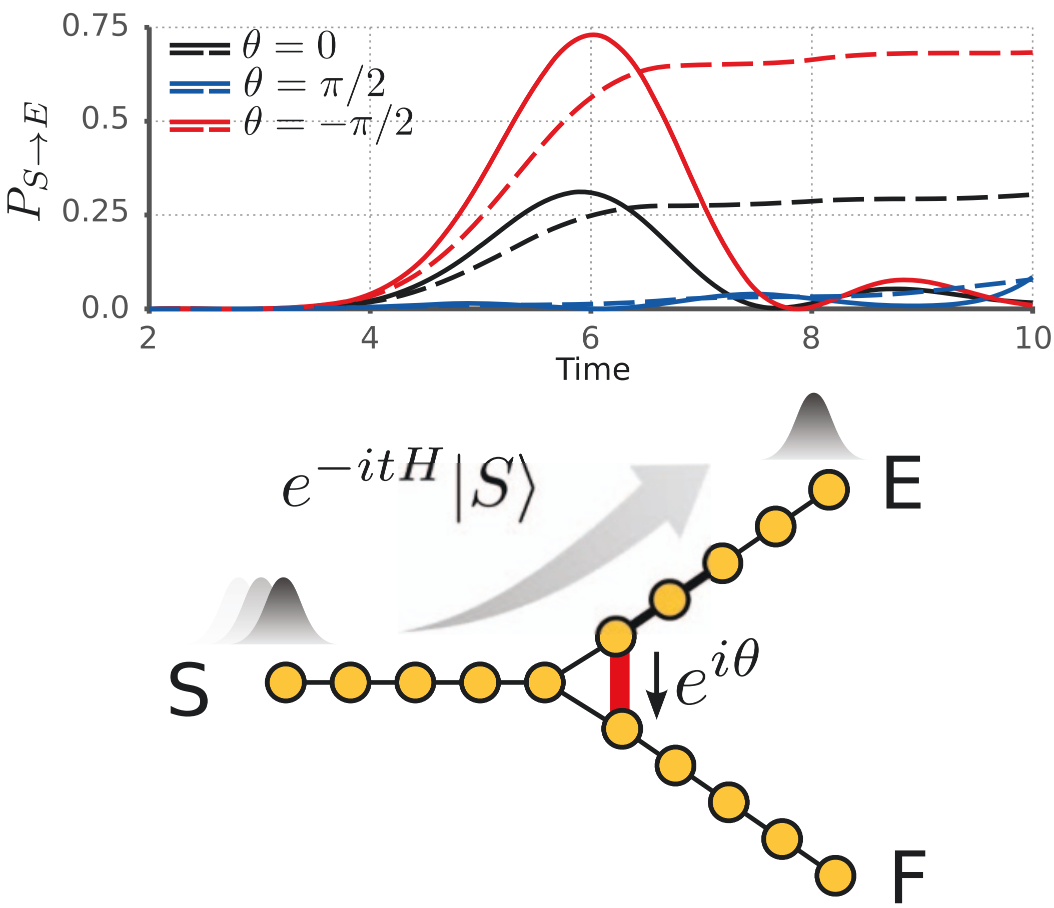


Figure 1 | The quantum switch. (a) Directional biasing: enhanced transport in the preferred direction. (b) The plot shows the occupancy probability $P_{S\to E}$ of site E with the particle initially starting from site S with and without a sink (dashed and solid lines, respectively). This evolution is time-reversal asymmetric as replacing t with $-t$ results in the particle moving from site S towards site F. When starting at site E, the particle evolves towards site F. By replacing t with $-t$, a particle initially at site E evolves towards the initial configuration (b). To recover time-reversal symmetric transition probabilities in the evolution (b), requires that one also performs the antiunitary operation[24] on the Hamiltonian mapping θ to $-\theta$. This has the same effect as reflecting the configuration horizontally across the page while leaving the site labels intact. chirality

antiunitary is like swapping the labels

Credits:
Massimo Frigerio, PhD Thesis
(2023)

6.3 Quasi-gauge transformations and phase configurations

Distinct configurations of phases on the edges of a graph $\mathcal{G}$, corresponding to different chiral Hamiltonians, can lead to the same transport probabilities between every pair of sites. We can therefore factor the space of all possible phase choices, and therefore the set of all Hamiltonians with the same underlying discrete structure and the same diagonal entries, into equivalence classes such that the Hamiltonians inside each class share the same site-to-site transition probabilities. It can be shown [331] that two chiral Hamiltonians H and H′ on the same graph $\mathcal{G}$ are equivalent in this sense if and only if they are related by a *quasi-gauge transformation*[1]:

equivalence class

$$\mathrm{H} \sim \mathrm{H}' \quad \Longleftrightarrow \quad \mathrm{H}' = \Lambda_{\vec{\alpha}} \mathrm{H} \Lambda_{\vec{\alpha}}^{\dagger} \tag{6.12}$$

link: diagonal unitary matrix

where $\Lambda_{\vec{\alpha}}$ is a diagonal unitary matrix with $(\Lambda_{\vec{\alpha}})_{jj} = e^{i\alpha_j}$ and $\vec{\alpha} \in [0, 2\pi)^N$. It is a simple task to show that

$$P_{j\to k}(t) = |\langle k|e^{-i\mathrm{H}t}|j\rangle|^2 = P'_{j\to k}(t) = |\langle k|e^{-i\mathrm{H}'t}|j\rangle|^2 \tag{6.13}$$

for $j, k = 1, ..., N$. Moreover, if we take generic pure states as the initial and final states, a quasi-gauge transformation will simply send a transition probability generated by H into *another* transition probability generated by H′, obtained just by application of the quasi-gauge transformation on the initial and final state. We have therefore the following result:

Theorem 6.3.1

Let H and H′ be two chiral Hamiltonians on the same graph $\mathcal{G}$ that are related via a quasi-gauge transformation. Then they will generate the *same* transition probabilities between any two localized states at all times. Moreover, the set of all transition probabilities generated by H for a time t, given *any pure initial and final state*, is a permutation of the same set generated by H′ for the same time t.

BTW localized sites

In this way, the phases of at most $N-1$ edges can be cancelled; indeed, if we multiply $\Lambda_{\vec{\alpha}}$ by an additional phase factor $e^{i\theta}$, the transformation in Eq.(6.12) applied to H

[1] Since these transformations do not affect the diagonal elements, H and H′ must already coincide on the diagonal.

N-1 free parameters in the diagonal unitary, neglecting the global phase will result in the same matrix H′, therefore only $N-1$ among the parameters of $\Lambda_{\vec{\alpha}}$ can matter. A tree graph with N vertices has exactly $N-1$ edges, and it is indeed the case that all chiral Hamiltonians on it belong to the same equivalence class of the adjacency matrix (disregarding the diagonal elements). In particular, on tree graphs the properties in Eq.(5.32) and Eq.(5.33) hold also for chiral quantum walks. However, as soon as the number of edges is larger than $N-1$, there will be at least one loop in the graph, and there will be a phase that cannot be cancelled. This implies different values of the transition probabilities, but *not necessarily* the breaking of the two symmetries. Indeed, the following result holds true:

trees again

when loops appear

Theorem 6.3.2

Let $\mathcal{G}$ be a bipartite graph. Then the time-reversal symmetry is preserved by any adjacency-type chiral quantum walk on $\mathcal{G}$. The same holds true for the spatial symmetry, if present in the non-chiral case.

Proof. Let us prove that the time-reversal symmetry is preserved, since spatial symmetry immediately follows if the graph exhibits it. Let us name V_1 and V_2 the two partitions of the set of vertices of $\mathcal{G}$. If we apply a quasi-gauge transformations to all the vertices in V_1 with a phase of π, so that $|j\rangle \mapsto -|j\rangle$ for any $j \in V_1$, then, given that each edge in $\mathcal{G}$ is connected to exactly one vertex in V_1, all the off-diagonal elements of H reverse their sign, so H $\mapsto$ −H and time-reversal is realised by a quasi-gauge transformation, thus preserving the transition probabilities. they all change sign once □

total phases along each loop is invariant! important degree of freedom

It turns out that equivalence classes of chiral Hamiltonians can be characterized by the total phases along each loop of the graph [332], i.e. the product of the phase factors of all the edges involved in the loop, for a given orientation of it: two chiral Hamiltonians with the same diagonal entries are related by a quasi-gauge transformation if and only if all their loops total phases coincide. Loops phases are thus invariants for chiral Hamiltonians, and they will be their central degrees of freedom.

In general, given the graph $\mathcal{G} = (V, E)$ and denoting by $|E|$ its number of edges and by $|V|$ its number of vertices, the actual number of relevant phase parameters for transport will be $|E|-|N|+1$, since each edge can bear a phase and $N-1$ of them are arbitrary under quasi-gauge transformations. For planar graphs, the corresponding independent loops are easily identified, since they are related to the number of regions in which the plane is partitioned by the graph when it is embedded in it without self-intersections. Letting F be this number of *finite* regions[2], thanks to the Euler characteristic of planar graphs we have: planar graphs

e.g.

$$|V| - |E| + F = 1 \quad \Rightarrow \quad F = |E| - |V| + 1 \tag{6.14}$$

therefore F is indeed equal to the number of relevant phase parameters.

6.4 Chiral quantum walks on lattices as discretized gauge theories

To provide a physical intuition for the phase degrees of freedom, we remark that the idea of amending for a local change of phase in the wave-function by changing some parameters in the Hamiltonian reminds of local gauge invariance with gauge group $U(1)$, i.e. intuition

[2] In the mathematical literature it is common to include also the outside, infinite region when counting faces of planar graphs. To avoid confusions, we specify that we count only finite regions.

coupling

of a coupling between a charged particle described by the wavefunction and a classical abelian gauge field. Indeed, we will now argue that the most general unweighted chiral CTQW on a regular lattice and in any dimension may be interpreted as the discretization of the non-relativistic Schrödinger equation for a scalar charged particle coupled to a classical electromagnetic field. Let us consider an infinite, regular d-dimensional lattice (not necessarily cubic). We require that all the edges have the same weight, reflecting the idea that, apart for the effect of potentials, the transition amplitudes between nearby points in an empty Euclidean space should just depend upon the absolute distance between them. We will refer to a chiral, unweighted QW on a regular lattice as a *homogeneous* chiral CTQW, and these are our natural candidates to be interpreted as spatial discretizations of quantum theories in Euclidean spaces.

unweighted chiral CTQW on regular lattice (not regular graph!)

homogeneous

Let us start by considering the simple scenario of an infinite 2D square lattice. Referring to some arbitrary vertex, we can label each point with a pair of integers $(n, m) \in \mathbb{Z}^2$ and the corresponding localized state will be $|n, m\rangle$. We will denote by $\Psi_{n,m}(t) = \langle n, m|\Psi(t)\rangle$ the amplitude of the walker at time t to be found at site (n, m), and by $\Psi(t)$ the vector of the amplitudes over all the sites of the lattice. It is also convenient to explicitly write the rate γ that governs the time evolution, so that the Schrödinger equation would be symbolically written as ($\hbar = 1$)

sites and projection

$$i\partial_t\vec{\Psi} \;=\; \gamma \mathrm{H}\cdot\vec{\Psi} \tag{6.15}$$

The most general Hamiltonian matrix H for a CTQW on this lattice will be an infinite, sparse matrix specified by the following entries [3]:

$$\begin{aligned}\frac{1}{\gamma}\langle n', m'|\mathrm{H}|n, m\rangle \;=\;& [4 + d(n, m)]\,\delta_{n',n}\delta_{m',m} + \\ & -\; e^{if_x(n'-1,m')}\delta_{n',n+1}\delta_{m',m} \;-\; e^{-if_x(n',m')}\delta_{n',n-1}\delta_{m',m} + \\ & -\; e^{if_y(n',m'-1)}\delta_{n',n}\delta_{m',m+1} \;-\; e^{-if_y(n',m')}\delta_{n',n}\delta_{m',m-1}\end{aligned} \tag{6.16}$$

dettano la legge per avanzare di fase con le celle

first neighbors

where $\delta_{p,n}$ is the Kronecker delta and $f_x(n, m)$, $f_y(n, m)$ and $d(n, m)$ are real valued functions of the nodes coordinates. Now let a be the lattice spacing, which can be assumed to be the same in all directions without loss of generality. By exploring all possible scaling laws of the functions f_x, f_y and d with a, a slight variation of an argument by Feynman (see the last Chapter of [346]) shows that the only nontrivial continuous limit ($a \to 0$) of this model leads to an Hamiltonian operator $\hat{\mathrm{H}}_c(\vec{x})$ of the following form:

continuous limit

$$\hat{\mathrm{H}}_c(\vec{x})\Psi(\vec{x}) \;=\; -K\left(\nabla - i\vec{F}(\vec{x})\right)^2\Psi(\vec{x}) + U(\vec{x})\Psi(\vec{x}) \tag{6.17}$$

where now:

$$\vec{x} \in \mathbb{R}^2\,, \quad U(\vec{x}) = \gamma d(\vec{x})\,, \quad K = \lim_{a\to 0} a^2\gamma \tag{6.18}$$

$$\vec{F}(\vec{x}) = \lim_{a\to 0}\frac{1}{a}\left(f_x(\vec{x}), f_y(\vec{x})\right) \tag{6.19}$$

and $d(\vec{x})$, $f_x(\vec{x})$, $f_y(\vec{x})$ are the continuum generalizations of $d(n, m)$, $f_x(n, m)$ and $f_y(n, m)$, respectively.

[3] Since the functions $f_x(n, m)$, $f_y(n, m)$ and $d(n, m)$ are completely unconstrained, this is indeed the most general Hamiltonian with nearest neighbor interactions on a square 2D lattice.

PROOF

To prove this, let us relabel the lattice points as $(n, m) \mapsto (x_n, y_m)$ where $x_n = na$ and $y_m = ma$. By considering f_y and f_x to be independent functions, we can indeed take the same lattice spacing along x and along y, without loss of generality. Now we imagine to extend Ψ to a smooth complex-valued function on the whole $\mathbb{R}^2$ plane, that interpolates between the values of the amplitude at the lattice sites and do the same for the real valued functions $d(x, y)$, $f_x(x, y)$ and $f_y(x, y)$. With this notation, the action of the Hamiltonian on Ψ could be written as: grating step

$$\begin{aligned} \frac{1}{\gamma}\left[\mathrm{H}\cdot\Psi\right](x,y) \;=\; & \left[4 + d(x,y)\right]\Psi(x,y) + \\ & -\; e^{i f_x(x-a,y)}\Psi(x-a,y) \;-\; e^{-i f_x(x,y)}\Psi(x+a,y) + \\ & -\; e^{i f_y(x,y-a)}\Psi(x,y-a) \;-\; e^{-i f_y(x,y)}\Psi(x,y+a). \end{aligned} \tag{6.20}$$

First, let us focus on the terms with translations along the x direction, the second line of Eq.(6.20). Since we have to *define* a continuum limit, we can freely postulate the scaling of f_x with a. If we take f_x to be $O(1)$, by expanding the second line up to first order in a we find:

suppose one scaling

$$\begin{aligned} & -e^{i f_x} e^{-i a \partial_x f_x}\left[\Psi - a\partial_x\Psi\right] - e^{i f_x}\left[\Psi + a\partial_x\Psi\right] + O(a^2) \;= \\ & = \; -2\left(\cos f_x\right)\Psi + a e^{i f_x}\left(\partial_x f_x\right)\Psi + 2ia\left(\sin f_x\right)\partial_x\Psi + O(a^2) \end{aligned} \tag{6.21}$$

diagonal correction to the scalar potential

where we wrote $f_x \equiv f_x(x, y)$ and $\Psi \equiv \Psi(x, y)$ for shortness. Taking the limit $a \to 0$ yields a contribution which is diagonal in the position, so it is just a correction to the scalar potential $d(x, y)$. Analogously, if we expanded the last line assuming f_y of $O(1)$ in a, we would have found a correction term to $d(x, y)$ of the form $-2\cos f_y$. Now suppose that f_x is $O(a)$, so that we can write $f_x(x, y) = aF(x, y)$. Expanding again the second line of Eq.(6.20), this time up to $O(a^2)$, we find:

suppose a different scaling

$$\begin{aligned} & -\;\left(1 + iaF_x - ia^2\partial_x F_x - \frac{1}{2}a^2F_x^2\right)\left(\Psi - a\partial_x\Psi + \frac{1}{2}a^2\partial_x^2\Psi\right) + \\ & -\;\left(1 - iaF_x - \frac{1}{2}a^2F_x^2\right)\left(\Psi + a\partial_x\Psi + \frac{1}{2}a^2\partial_x^2\Psi\right) + O(a^3) \;= \\ & = \; -2\Psi - a^2\Big(\partial_x - iF_x\Big)^2\Psi \;+\; O(a^3) \end{aligned} \tag{6.22}$$

and $F_x \equiv F_x(x, y)$. Doing the same for the third line of Eq.(6.20), with $f_y = aF_y$, the net result is:

$$\left[\mathrm{H}\cdot\Psi\right](x,y) \;=\; -\gamma a^2\left(\nabla - i\vec{F}(x,y)\right)^2\Psi(x,y) \;+\; \gamma d(x,y)\Psi(x,y) \;+\; O(a^3) \tag{6.23}$$

take infinite gamma

where $\vec{F} = (F_x, F_y)$. If we naively take $a \to 0$ at this point, the result would again be trivial. However, unlike the previous case, here we can take $\gamma = K/a^2$, with $K \sim O(1)$ and constant, and also $\lim_{a\to 0}\gamma d(x, y) = U(x, y)$, where we are forced to assume $d(x, y) \sim O(a^2)$ to get a finite continuum limit with a potential field landscape. For these choices, the continuum limit is no longer trivial and we arrive at the desired result: take small d

$$\lim_{a\to 0}\;\left[\mathrm{H}\cdot\Psi\right](x,y) \;=\; -K\left(\nabla - i\vec{F}(x,y)\right)^2\Psi(x,y) \;+\; U(x,y)\Psi(x,y) \tag{6.24}$$

a posteriori, the scaling was ideal

We also remark that taking $f_x \sim O(1)$ and $f_y \sim O(a)$, or vice versa, would prevent γ to be $O(a^{-2})$, otherwise the $-2\gamma \cos f_x$ term would diverge as $a \to 0$. Therefore, a mixed choice of scaling laws for f_x and f_y yields a diagonal action of H on Ψ, hence a trivial result. The last option is to postulate $f_x \sim O(a^{2+\epsilon})$ for $\epsilon > 0$. The only option to find a nontrivial continuum limit is again for $\gamma = K/a^2$ and the result is the same as neglecting f_x altogether to begin with. We have thus proved that the only nontrivial continuum limit of a generic chiral CTQW on a square 2D lattice is given by Eq.(6.24). To conclude, let us highlight the fact that the scaling laws $f_x \sim O(a)$ and $f_y \sim O(a)$ are, in a sense, the most natural choice: these functions define the phases for the hopping amplitudes between neighbors on the lattice, and it is reasonable to require that these phases should go to 0 as the distance between neighboring points is reduced.

there are no more terms to include

phases are connected with distance

We have thus seen that the existence of all these limits is a prerequisite to find a nontrivial theory as $a \to 0$. We can now restore $\hbar$, write $U(x) = qV(x)$, $F(x) = qA(x)$ and $K = \frac{\hbar^2}{2m}$ in order to arrive at the standard nonrelativistic Schrödinger equation for a scalar particle with mass m and charge q in the presence of an electromagnetic field:

$$i\hbar \frac{\partial}{\partial t}\Psi(\vec{x}) = -\frac{\hbar^2}{2m}\left(\nabla - iq\vec{A}(\vec{x})\right)^2 \Psi(\vec{x}) + qV(\vec{x})\Psi(\vec{x}) \tag{6.25}$$

where $\vec{A}(\vec{x})$ is the vector potential and $V(\vec{x})$ is the scalar potential. The derivation can be readily generalized to cubic lattices and, even more generally, to regular lattices in any dimension by introducing the appropriate vector potential $\vec{A}(x)$ and with suitable discretizations of the Laplacian [258], showing that homogeneous chiral CTQWs, despite being the most general of their kind, are in fact always equivalent to discretizations of the Schrödinger equation for a scalar particle in the presence of an electromagnetic field, also suggesting a practical implementation for chiral CTQWs, especially with *synthetic* gauge fields [338]. Notice that the reasoning can also be inverted to deduce the form of the phases $f(n, m)$ from a given vector potential. The result is known as Peierls substitution